Lab 9: Logistic and Count-Variable Regression

Analysis of Logistic Regression and Poisson Regression Results

This report summarizes the Python output for two datasets: birthwt.csv for logistic regression and places_2025.csv for Poisson regression. It uses reported model summaries, classification metrics, p-values, and variance inflation factors (VIF).

1. Logistic Regression

The logistic regression model used low as the binary response and smoke, ht, and ui as predictors. There were 189 complete observations. The results of logistic regression analysis showed convergence with a likehood-ratio test being statistically significant (LLR p-value = 0.002125), indicating that all predictors collectively provide better fit than the intercept-only model.

Measure	Result
Test-set accuracy	68.42%
Pseudo-R^2	0.0625
Likelihood-ratio test p-value	0.002125
Complete observations	189

Coefficients that are statistically significant and positive include smoke, which is 0.6831 (p-value of 0.038), ht (1.4174 and p = 0.023), and ui (1.0384 and p = 0.015). In this case, when other predictors are held constant, high values of these predictors result in higher log-odds of low = 1. These are correlations, not evidence of cause and effect.

The confusion matrix was [[25, 1], [11, 1]]. The model accurately predicted 25 out of 26 instances of class 0, while only predicting 1 out of 12 instances of class 1. The recall score for class 1 was 0.08, so the model failed to predict most positive instances. Thus, the accuracy of 68.42% is not indicative of good performance in terms of predicting class 1.

2. Poisson (Count-Variable) Regression

The Poisson model used mhlth_crudeprev as the dependent variable while there were 11 independent variables that came from places_2025.csv. There were 2,299 cases for which the data was complete and there was convergence of the model. The test of Likelihood ratio was statistically significant ($p = 1.908 \times 10nnn$), showing that the model fits better than the intercept-only model using this test.

Four predictors were statistically significant at the 0.05 significance level:

Predictor	Coefficient	p-value	Direction
depression_crudeprev	0.0436	< 0.001	Positive
selfcare_crudeprev	-0.0733	0.007	Negative
indeplive_crudeprev	0.1032	< 0.001	Positive
loneliness_crudeprev	0.0345	< 0.001	Positive

The remaining predictors—binge_crudeprev, checkup_crudeprev, csmoking_crudeprev, obesity_crudeprev, sleep_crudeprev, foodinsecu_crudeprev, and housinsecu_crudeprev are other predictors that did not satisfy the $p < 0.05$ threshold. The coefficients are conditional associations; interpretation should be cautious because predictors may be correlated.

3. Model Diagnostics and Limitations

The output reports that the response variable includes non-integer values. Given the name of mhlth_crudeprev, it can be seen that the model estimates a measure of crude prevalence rather than a count value. As Poisson regression is applied to count data, the choice of the model needs to be examined prior to the interpretation of the results. A model designed for rates or proportions may be more appropriate, depending on the dataset definition and available denominator information.

Multicollinearity is indicated by the VIF values for the variables foodinsecu_crudeprev (62.83), housinsecu_crudeprev (46.84), indeplive_crudeprev (30.95), and selfcare_crudeprev (30.51).This implies there is some form of overlap between the predictors, and hence the coefficients may not be stable or even the p-values interpretable. Further analysis of the predictor relationships may be required.

While the logistic classifier can accurately predict class 0 but detects a small number of cases in class 1. In certain scenarios, class balance, recall, and threshold require additional analysis. The presented performance measures are based on one train/test split

4. Conclusion

Both scripts ran successfully and created model output. The variables in the logistic model that were found to be statistically significant are smoke, ht, and ui, though the recall for class 1 is low. For the Poisson model, depression, selfcare, indeplive, and loneliness are the variables that are found to be statistically significant at the 0.05 level. Nevertheless, the response variable for the Poisson model is the prevalence/rate and not the counts, while some of the variables have high VIF values. It is important to note these issues before coming up with any conclusions.